

From the Dirichlet–Neumann Operator to the Nonlinear Schrödinger Equation: Deep-Water Wave Regimes and the Benjamin–Feir Instability

Notes for DNO–FNO project

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1 The Water Wave Problem via the DNO

Consider irrotational, incompressible potential flow in the domain

$$\Omega(t) = \{(x, y) \in \mathbb{R} \times \mathbb{R} : -h < y < \eta(x, t)\},$$

with free surface $\eta(x, t)$ and flat bottom at $y = -h$. The velocity potential $\phi(x, y, t)$ satisfies

$$\Delta\phi = 0 \quad \text{in } \Omega(t), \quad \left. \frac{\partial\phi}{\partial y} \right|_{y=-h} = 0.$$

This is a free boundary problem: the domain depends on the solution.

1.1 Zakharov–Craig–Sulem Formulation

The key reduction (Zakharov [1], Craig & Sulem [2]) reformulates the problem entirely on the free surface. Define the surface potential $\xi(x, t) = \phi(x, \eta(x, t), t)$ and the **Dirichlet–Neumann operator**:

$$G(\eta)\xi = \phi_y|_{y=\eta} - \eta_x \phi_x|_{y=\eta}.$$

This is the standard Dirichlet-to-Neumann map for the Laplacian on $\Omega(t)$, weighted by the surface geometry. The Euler equations then reduce to:

$$\eta_t = G(\eta)\xi, \tag{1}$$

$$\xi_t = -g\eta - \frac{1}{2}|\xi_x|^2 + \frac{(G(\eta)\xi + \eta_x \xi_x)^2}{2(1 + |\eta_x|^2)}. \tag{2}$$

This is a Hamiltonian system with $H = \frac{1}{2} \int [\xi G(\eta)\xi + g\eta^2] dx$ and canonically conjugate variables (η, ξ) .

The payoff is dimensional reduction: instead of solving a 2D elliptic problem at every time step, one evaluates $G(\eta)$ acting on a function of one spatial variable.

1.2 Taylor Expansion of the DNO

$G(\eta)$ is analytic in η for sufficiently smooth surfaces (Coifman–Meyer, Craig–Nicholls), with convergent Taylor expansion:

$$G(\eta) = \sum_{j=0}^{\infty} G_j(\eta), \quad G_j \text{ homogeneous of degree } j \text{ in } \eta.$$

The zeroth-order term is the Fourier multiplier

$$\widehat{G_0\xi}(k) = |k| \tanh(h|k|) \hat{\xi}(k). \tag{3}$$

Higher terms follow from a recursion:

$$G_1(\eta) = D\eta D - G_0 \eta G_0, \tag{4}$$

$$G_2(\eta) = -\frac{1}{2}(G_0 \eta^2 D^2 + D^2 \eta^2 G_0 - 2G_0 \eta G_0 \eta G_0), \tag{5}$$

where $D = -i\partial_x$.

Depth dependence. The symbol $|k| \tanh(h|k|)$ interpolates between two regimes:

- Finite depth ($h \sim 1$): G_0 damps high modes.
- Deep water ($h \rightarrow \infty$): $G_0 \rightarrow |D|$, the half-derivative.

Since G_0 appears in the recursion for every G_j , changing h changes the entire operator.

Numerical truncation. In practice, $G(\eta) \approx \sum_{j=0}^M G_j(\eta)$ with $M = 4\text{--}6$. Each G_j is computed via FFTs at cost $O(M^2 N \log N)$. Convergence is exponential in M for smooth η [7].

2 Derivation of the NLS

The **nonlinear Schrödinger equation** (NLS) is the dispersive PDE

$$iA_T + p A_{XX} + q |A|^2 A = 0, \quad (6)$$

where $A(X, T) \in \mathbb{C}$ is the envelope of a modulated wave train and $p, q \in \mathbb{R}$ are coefficients determined by the underlying physics. When $pq > 0$ the equation is *focusing* (admits solitons, exhibits modulational instability); when $pq < 0$ it is *defocusing*.

In this section we derive (6) from the water wave system (1)–(2) in the deep-water limit $G_0 = |D|$, identifying p and q explicitly in terms of the carrier wavenumber k_0 and the DNO. The small parameter is the wave steepness $\varepsilon = k_0 a_0$. We write $E = e^{i\theta}$ throughout.

2.1 Motivation

The system (1)–(2) is exact but expensive to simulate and difficult to analyze directly. In deep water, the dominant observed structures are *modulated wave trains* — a fast carrier oscillation whose amplitude and phase vary slowly in space and time. The NLS captures the leading-order dynamics of this slow modulation, reducing the full infinite-dimensional system to (6).

This has both analytical and practical consequences. Analytically, the NLS is integrable (solvable by inverse scattering), so its solution space is completely understood: solitons, breathers, modulational instabilities all have exact descriptions. Practically, the NLS tells us *which families of surface states* (η, ξ) *are physically relevant* in deep water — and hence what training data a neural operator should see (Section 6).

2.2 The Linear Problem

Linearize (1)–(2) around rest ($\eta = \xi = 0$, $G(\eta) \approx G_0$): $\eta_t = G_0 \xi$, $\xi_t = -g\eta$. Plane waves $\propto e^{i(kx - \omega t)}$ satisfy $\omega^2 = g|k|$.

A narrow-band wave packet centered at $k_0 > 0$ has Fourier support near k_0 :

$$\eta(x, t) = \int_{\mathbb{R}} \hat{f}(k - k_0) e^{i(kx - \omega(k)t)} dk = \left(\int \hat{f}(\varepsilon \kappa) e^{i\varepsilon \kappa (x - \omega'(k_0)t) + i\frac{\varepsilon^2}{2} \omega''(k_0) \kappa^2 t + \dots} \frac{d\kappa}{\varepsilon} \right) e^{i(k_0 x - \omega_0 t)}.$$

The integral in parentheses is the slowly varying complex envelope $A(X, T)$, with $X = \varepsilon(x - c_g t)$ and $T = \varepsilon^2 t$. Its modulus $|A|$ is the local wave amplitude; $\arg(A)$ is the local phase correction to the carrier $E = e^{i\theta}$.

2.3 Multiple-Scales Setup

Fast and slow variables. Carrier phase $\theta = k_0 x - \omega_0 t$ with $\omega_0 = \sqrt{gk_0}$. Slow variables $X = \varepsilon(x - c_g t)$, $T = \varepsilon^2 t$, with group velocity $c_g = \omega'(k_0) = \omega_0/(2k_0)$. Derivatives become:

$$\partial_x = k_0 \partial_\theta + \varepsilon \partial_X, \quad \partial_t = -\omega_0 \partial_\theta - \varepsilon c_g \partial_X + \varepsilon^2 \partial_T. \quad (7)$$

G_0 on modulated harmonics. For $k_0 > 0$, the symbol $|k| = k$ for k near k_0 (exactly, not approximately). So for any slowly modulated harmonic $f(X) E^n$ with $n \geq 1$:

$$G_0[f(X) E^n] = (nk_0 - i\varepsilon \partial_X) f \cdot E^n. \quad (8)$$

This is *exact* in deep water — there are no $O(\varepsilon^2)$ corrections from G_0 alone, because $|k|$ is linear for $k > 0$. (At finite depth, $|k| \tanh(h|k|)$ is not linear, and its curvature would contribute additional terms here.)

Ansatz. Expand $\eta = \varepsilon \eta_1 + \varepsilon^2 \eta_2 + \varepsilon^3 \eta_3 + \dots$ and $\xi = \varepsilon \xi_1 + \varepsilon^2 \xi_2 + \varepsilon^3 \xi_3 + \dots$, with

$$\eta_1 = AE + \bar{A}E^{-1}, \quad \xi_1 = BE + \bar{B}E^{-1}, \quad (9)$$

where $B(X, T)$ is to be determined in terms of A . The corrections η_j, ξ_j for $j \geq 2$ are expanded in Fourier harmonics: $\eta_j = \sum_n \eta_j^{(n)}(X, T) E^n$.

2.4 $O(\varepsilon)$: Dispersion Relation

Apply (7) to $\eta_t = G(\eta)\xi$ at $O(\varepsilon)$, coefficient of E :

$$-i\omega_0 A = k_0 B \quad \implies \quad B = \frac{-i\omega_0}{k_0} A.$$

From $\xi_t = -g\eta$ at $O(\varepsilon)$, coefficient of E :

$$-i\omega_0 B = -gA \quad \implies \quad B = \frac{ig}{\omega_0} A = \frac{-i\omega_0}{k_0} A,$$

where the last equality uses $g = \omega_0^2/k_0$. Both equations give the same B ; consistency requires $\omega_0^2 = gk_0$. $\square$

2.5 $O(\varepsilon^2)$: Group Velocity and Forced Harmonics

At this order, the slow X -derivatives of A, B enter, and the quadratic nonlinearities in (2) generate forcing at harmonics E^0 (mean) and E^2 (second harmonic).

2.5.1 No secularity at E^1

The system at the fundamental E^1 reads (kinematic, then dynamic):

$$-i\omega_0 \eta_2^{(1)} - c_g A_X = k_0 \xi_2^{(1)} + (-iB_X), \quad (10)$$

$$-i\omega_0 \xi_2^{(1)} - c_g B_X = -g\eta_2^{(1)}. \quad (11)$$

The $-iB_X$ in (10) comes from the $\varepsilon\partial_X$ correction in (8); the $-c_g A_X$ and $-c_g B_X$ come from $\varepsilon c_g \partial_X$ in ∂_t . (No nonlinear terms at E^1 : the quadratic products $\xi_{1,x}^2$ and $(G_0\xi_1)^2$ produce only E^0 and E^2 .)

Equations (10)–(11) are two equations for $\eta_2^{(1)}$ and $\xi_2^{(1)}$, but the coefficient matrix $\begin{pmatrix} -i\omega_0 & -k_0 \\ -g & i\omega_0 \end{pmatrix}$ is singular ($\omega_0^2 = gk_0$). A solution exists iff the forcing satisfies a solvability condition. Substituting $B = -i\omega_0 A/k_0$ and $c_g = \omega_0/(2k_0)$ into the forcing terms, one checks directly that the A_X terms cancel. So there is no secularity at E^1 — the group velocity choice $c_g = \omega'(k_0)$ guarantees this. We set $\eta_2^{(1)} = \xi_2^{(1)} = 0$ (absorbed into A by redefinition).

2.5.2 Second harmonic E^2

The quadratic nonlinearities contribute at E^2 . Compute the relevant products at leading order, using $B = -i\omega_0 A/k_0$:

$$\xi_{1,x} = ik_0 B E - ik_0 \bar{B} E^{-1}, \quad (12)$$

$$\xi_{1,x}^2 = -k_0^2 (B^2 E^2 - 2|B|^2 + \bar{B}^2 E^{-2}), \quad (13)$$

$$(G_0\xi_1)^2 = k_0^2 (B^2 E^2 + 2|B|^2 + \bar{B}^2 E^{-2}). \quad (14)$$

The combined nonlinear forcing in (2) at E^2 is

$$\left[-\frac{1}{2}\xi_{1,x}^2 + \frac{1}{2}(G_0\xi_1)^2\right]_{E^2} = k_0^2 B^2 = -\omega_0^2 A^2.$$

(The numerator $(G\xi + \eta_x \xi_x)^2$ reduces to $(G_0\xi_1)^2$ at this order because $\eta_x \xi_x$ is $O(\varepsilon^2)$ but only has E^0 and E^2 — it contributes the same way.)

The kinematic and dynamic equations at E^2 give:

$$-2i\omega_0 \eta_2^{(2)} = 2k_0 \xi_2^{(2)}, \quad (15)$$

$$-2i\omega_0 \xi_2^{(2)} = -g\eta_2^{(2)} - \omega_0^2 A^2. \quad (16)$$

From (15): $\xi_2^{(2)} = -i\omega_0 \eta_2^{(2)}/k_0$. Substitute into (16):

$$-2i\omega_0 \cdot \frac{-i\omega_0 \eta_2^{(2)}}{k_0} = -g\eta_2^{(2)} - \omega_0^2 A^2 \quad \implies \quad -\frac{2\omega_0^2}{k_0} \eta_2^{(2)} = -g\eta_2^{(2)} - \omega_0^2 A^2.$$

Using $g = \omega_0^2/k_0$: the left side is $-2g\eta_2^{(2)}$, the right is $-g\eta_2^{(2)} - \omega_0^2 A^2$, giving

$$\boxed{\eta_2^{(2)} = k_0 A^2}, \quad \xi_2^{(2)} = \frac{-i\omega_0}{k_0} \cdot k_0 A^2 = -i\omega_0 A^2. \quad (17)$$

2.5.3 Mean flow E^0

The E^0 nonlinear forcing in the dynamic equation is

$$\left[-\frac{1}{2}\xi_{1,x}^2 + \frac{1}{2}(G_0\xi_1)^2\right]_{E^0} = k_0^2 |B|^2 - k_0^2 |\bar{B}|^2 = 0.$$

This cancellation is specific to deep water (at finite depth, G_0 on E^{+1} and E^{-1} gives different symbols, and the mean flow is nonzero). Therefore:

$$\eta_2^{(0)} = 0, \quad \xi_2^{(0)} = 0 \quad (\text{deep water}). \quad (18)$$

2.6 $O(\varepsilon^3)$: The NLS

2.6.1 Solvability framework

At $O(\varepsilon^3)$, the system at E^1 has the same singular matrix as before:

$$-i\omega_0\eta_3^{(1)} - k_0\xi_3^{(1)} = R_K, \quad (19)$$

$$-g\eta_3^{(1)} + i\omega_0\xi_3^{(1)} = R_D, \quad (20)$$

where R_K, R_D collect all $O(\varepsilon^3)$, E^1 terms not involving $\eta_3^{(1)}$ or $\xi_3^{(1)}$. The solvability condition (left null vector of the coefficient matrix) is:

$$i\omega_0 R_K + k_0 R_D = 0. \quad (21)$$

We now compute R_K and R_D .

2.6.2 Dispersive contribution (the $p A_{XX}$ term)

The dispersion comes from the interplay of ∂_t^2 and G_0 across the kinematic and dynamic equations, not from G_0 alone. The cleanest way to see this: differentiate $\eta_t = G_0\xi + \dots$ in time to get $\eta_{tt} = -gG_0\eta + \dots$, i.e. $(\partial_t^2 + gG_0)\eta = \text{nonlinear forcing}$.

Acting on $\varepsilon A(X, T) E$:

$$\partial_t(\varepsilon A E) = \varepsilon(-i\omega_0 A - \varepsilon c_g A_X + \varepsilon^2 A_T) E, \quad (22)$$

$$\partial_t^2(\varepsilon A E) = \varepsilon(-\omega_0^2 A + 2i\omega_0 \varepsilon c_g A_X + \varepsilon^2(c_g^2 A_{XX} - 2i\omega_0 A_T)) E, \quad (23)$$

$$gG_0(\varepsilon A E) = \varepsilon(gk_0 A - ig\varepsilon A_X) E = \varepsilon(\omega_0^2 A - i\varepsilon\omega_0^2 A_X/k_0) E. \quad (24)$$

Adding:

$$(\partial_t^2 + gG_0)[\varepsilon A E] = \varepsilon^3(c_g^2 A_{XX} - 2i\omega_0 A_T) E$$

— the $O(\varepsilon)$ and $O(\varepsilon^2)$ terms cancel (using $\omega_0^2 = gk_0$ and $c_g = \omega_0/(2k_0)$).

So the linear part of the solvability condition at $O(\varepsilon^3)$ contributes $c_g^2 A_{XX} - 2i\omega_0 A_T$. Rearranging:

$$iA_T = \frac{c_g^2}{2\omega_0} A_{XX} + \dots = \frac{\omega_0}{8k_0^2} A_{XX} + \dots$$

This gives the dispersive coefficient:

$$p = \frac{c_g^2}{2\omega_0} = \frac{\omega_0}{8k_0^2}. \quad (25)$$

2.6.3 Nonlinear contribution (the $q|A|^2 A$ term)

The nonlinear forcing at $O(\varepsilon^3)$, E^1 has several sources. We enumerate them, noting that $\eta_2^{(0)} = 0$ (deep water) eliminates several terms.

(a) Quadratic terms in (2) coupling η_1 with η_2 . The products $\xi_{1,x} \cdot \xi_{2,x}$ and $G_0\xi_1 \cdot G_0\xi_2$ at E^1 involve $\xi_1 (E^{\pm 1})$ interacting with $\xi_2^{(2)} E^2$:

$$[\xi_{1,x} \cdot \xi_{2,x}]_{E^1} = (ik_0 \bar{B})(-2ik_0)(-i\omega_0 A^2) + (\text{from } E^{-1} \cdot E^2), \quad (26)$$

$$[G_0\xi_1 \cdot G_0\xi_2]_{E^1} = (k_0 \bar{B})(2k_0)(-i\omega_0 A^2). \quad (27)$$

(b) $G_1(\eta_2)\xi_1$ at E^1 . $G_1(\eta)\xi = D(\eta D\xi) - G_0(\eta G_0\xi)$. With $\eta_2^{(2)}E^2$ and $\xi_1 = BE + \bar{B}E^{-1}$: the E^1 component comes from $E^2 \cdot E^{-1}$, giving terms proportional to $\eta_2^{(2)}\bar{B} = k_0A^2 \cdot (i\omega_0\bar{A}/k_0)$.

(c) $G_1(\eta_1)\xi_2$ at E^1 . Similarly, $\eta_1 = AE + \bar{A}E^{-1}$ with $\xi_2^{(2)}E^2$: the E^1 component comes from $E^{-1} \cdot E^2$, proportional to $\bar{A} \cdot \xi_2^{(2)} = \bar{A}(-i\omega_0A^2)$.

(d) $G_2(\eta_1)\xi_1$ at E^1 . This involves η_1^2 (which has E^0 and $E^{\pm 2}$ components) acting on ξ_1 through the second-order DNO correction. The E^1 component comes from the E^0 part of η_1^2 ($= 2|A|^2$) times the E^1 part of ξ_1 .

Each of (a)–(d) produces a term proportional to $|A|^2A \cdot E$. Collecting all contributions and applying the solvability condition (21), the nonlinear coefficient works out to:

$$q = \frac{\omega_0 k_0^2}{2}. \quad (28)$$

(The detailed algebra is carried out in, e.g., Mei, Stiassnie & Yue [13], §13.3, or Ablowitz [14], §6.4.)

2.6.4 The NLS

Combining the dispersive and nonlinear contributions, the solvability condition at $O(\varepsilon^3)$ is

$$\boxed{iA_T + p A_{XX} + q |A|^2 A = 0} \quad (29)$$

with

$$p = \frac{\omega_0}{8k_0^2}, \quad q = \frac{\omega_0 k_0^2}{2}.$$

Since $pq > 0$ in deep water, the equation is **focusing**: it admits soliton solutions, and plane waves are modulationally unstable.

Rigorous justification. Düll, Schneider & Wayne [5] proved that NLS solutions approximate the full Euler equations on the natural $O(1/\varepsilon^2)$ time scale, with the key estimate being analyticity bounds on $G(\eta)$.

2.7 Finite Depth

For finite h , the same procedure applies but $G_0 = |D| \tanh(h|D|)$, so (8) acquires $O(\varepsilon^2)$ corrections from the curvature of the symbol. The dispersion relation becomes $\omega = \sqrt{gk \tanh(hk)}$, and critically, $\eta_2^{(0)} \neq 0$ — the mean flow no longer vanishes, modifying the nonlinear coefficient. The sign of q changes at $k_0 h \approx 1.363$: below this threshold, $pq < 0$ (defocusing), and plane waves are stable.

3 What the NLS–Euler Connection Buys Us

The derivation in Section 2 establishes that the NLS (29) governs the envelope dynamics of deep-water waves at $O(\varepsilon^3)$. This is useful in two distinct ways: the NLS gives us a *catalog of solutions*, and the Euler equations (via the DNO) give us the *exact nonlinear operator* we want to learn.

3.1 The NLS as a Solution Generator

The focusing NLS is integrable: its full solution space is known via inverse scattering. This gives closed-form expressions for envelope solitons, Akhmediev breathers, Peregrine breathers, and multi-soliton states — none of which are easy to guess from the Euler equations directly.

Each NLS solution $A(X, T)$ lifts to physical variables (η, ξ) via (9), producing an approximate solution of the full water wave equations valid to $O(\varepsilon^2)$. These lifted states serve as:

- **Initial conditions** for DNO-based time integration of the full Euler equations.
- **Standalone data points:** evaluate $G(\eta)\xi$ directly on the lifted (η, ξ) pair using the DNO series, without any time integration at all.

3.2 What Is Easy and What Is Not

Envelope solitons: easy. The NLS soliton (31) gives an explicit formula for A at any (X, T) . Lifting to (η, ξ) is a single evaluation of (9) — no time stepping, no iteration. One can generate thousands of soliton snapshots at different (k_0, A_0) and evaluate $G(\eta)\xi$ on each. The only cost is the DNO series evaluation itself.

Peregrine and Akhmediev breathers: easy as snapshots, interesting as trajectories. Like the soliton, the Peregrine (34) and Akhmediev (33) breathers have closed-form expressions. Generating (η, ξ) snapshots at any (X, T) is immediate. However, these are approximate solutions of Euler — the NLS truncates at $O(\varepsilon^3)$, so the lifted state drifts from the true Euler trajectory over time. If one wants data that reflects the *full* nonlinear dynamics (post-NLS corrections, recurrence breakdown, frequency downshift), one must time-integrate the Euler equations from the NLS-derived initial condition and snapshot along the trajectory.

Benjamin–Feir recurrence: requires time integration. The BF instability is easy to *set up* — perturb a Stokes wave with sidebands, which the NLS predicts will grow. But the interesting physics (recurrence, eventual breakdown) plays out over hundreds of carrier periods and involves the full nonlinear DNO. The NLS predicts exact recurrence; the Euler equations show it breaking down after 2–3 cycles [7]. Capturing this discrepancy requires running the DNO solver, which is expensive but produces the most physically rich training data.

3.3 Summary: Division of Labor

Role	NLS provides	Euler/DNO provides
Solution families	closed-form catalog	—
Initial conditions	(η, ξ) via lifting	—
$G(\eta)\xi$ evaluation	—	exact (to order M)
Long-time dynamics	approximate (integrable)	exact (non-integrable)
Post-NLS effects	—	recurrence breakdown, frequency downshift, breaking

In short: the NLS tells us *what to simulate*, and the DNO tells us *what the operator actually does* on those states. For operator learning, the NLS is a recipe book for generating diverse, physically motivated inputs (η, ξ) ; the DNO series provides the ground-truth labels $G(\eta)\xi$.

4 Exact Solutions of the Focusing NLS

The focusing NLS is integrable (inverse scattering), yielding several families of exact solutions that correspond to distinct deep-water wave phenomena. Each can be lifted to (η, ξ) via (9).

4.1 Plane Wave (Stokes Wave)

$$A(X, T) = A_0 e^{iqA_0^2 T}. \quad (30)$$

Spatially uniform carrier with nonlinear frequency correction. Starting point for stability analysis.

4.2 Envelope Soliton

$$A(X, T) = A_0 \operatorname{sech}\left(A_0 \sqrt{\frac{q}{2p}} X\right) \exp\left(\frac{i}{2} q A_0^2 T\right). \quad (31)$$

A localized wave packet propagating at c_g without dispersing — the deep-water counterpart of the KdV soliton, but modulating a carrier rather than forming a single hump. At $T = 0$:

$$\eta(x, 0) = 2\varepsilon A_0 \operatorname{sech}(\varepsilon A_0 \sqrt{q/(2p)} x) \cos(k_0 x). \quad (32)$$

Confirmed experimentally [11].

4.3 Akhmediev Breather

Spatially periodic, temporally localized [12]:

$$A(X, T) = A_0 \frac{\cosh(\sigma T - 2i\varphi) - \cos \varphi \cos(\kappa X)}{\cosh(\sigma T) - \cos \varphi \cos(\kappa X)} e^{iA_0^2 T}, \quad (33)$$

where $\cos \varphi = \kappa/(2A_0 \sqrt{q/p})$ and $\sigma = \kappa \sqrt{4qA_0^2/p - \kappa^2 q/p}$.

As $T \rightarrow -\infty$ this is the plane wave plus an infinitesimal $\cos(\kappa X)$ perturbation. At $T = 0$ the modulation reaches maximum amplitude. As $T \rightarrow +\infty$ it returns to the plane wave. This describes one complete cycle of the Benjamin–Feir instability (Section 5).

4.4 Peregrine Breather

The $\kappa \rightarrow 0$ limit of the Akhmediev breather [10]:

$$A(X, T) = A_0 \left(1 - \frac{4(1 + 2iqA_0^2 T)}{1 + 4q^2 A_0^4 T^2/p^2 + 2qA_0^2 X^2/p}\right) e^{iqA_0^2 T}. \quad (34)$$

Localized in both space and time, with maximum amplitude $3A_0$ — the prototypical rogue wave. Produces steep, localized surface states where $G_2, G_3, \dots$ are non-negligible.

5 The Benjamin–Feir Instability

Benjamin and Feir [9] showed that uniform deep-water Stokes waves are modulationally unstable. This is a consequence of the focusing character of the NLS.

5.1 Linear Stability Analysis

Perturb the plane wave at sideband wavenumbers $k_0 \pm \delta k$:

$$A = (A_0 + a_+ e^{i\delta k X + \Omega T} + a_- e^{-i\delta k X + \bar{\Omega} T}) e^{iqA_0^2 T}.$$

Linearizing (29):

$$\Omega^2 = p^2(\delta k)^2 [(\delta k)^2 - 2qA_0^2/p]. \quad (35)$$

Instability ($\Omega^2 < 0$) when

$$0 < |\delta k| < A_0 \sqrt{2q/p} = \sqrt{2} k_0^2 a_0. \quad (36)$$

Maximum growth rate $\Omega_{\max} = qA_0^2 = \omega_0 k_0^2 a_0^2/2$ at $|\delta k| = A_0 \sqrt{q/p}$. At steepness $k_0 a_0 = 0.13$, the e -folding time is ~ 19 carrier periods.

The coefficient q — and hence the instability bandwidth — originates from the nonlinear interaction mediated by $G_1(\eta)\xi$. The exact stability analysis (Berti, Maspero & Ventura [8]) uses the shape derivative of the full DNO.

5.2 Nonlinear Recurrence

The instability saturates. Energy cycles between carrier and sidebands in Fermi–Pasta–Ulam recurrence: growth $\rightarrow$ sideband maximum $\rightarrow$ return to carrier $\rightarrow$ repeat. The Akhmediev breather (33) is the exact NLS solution describing one cycle.

For the full Euler equations (via DNO), recurrence persists for 2–3 cycles before higher-order effects (absent at $O(\varepsilon^3)$) disrupt it.

5.3 DNO Simulation (Xu & Guyenne 2009)

Xu & Guyenne [7] simulated this with the full system (1)–(2), $M = 4$. Initial condition: Stokes wave ($k_1 = 9$, $k_1 a = 0.13$) with sideband perturbations at $k_l = 7$, $k_r = 11$:

$$\eta(x, 0) = \eta_0 + \varepsilon a \left[\cos(k_l x - \frac{\pi}{4}) + \cos(k_r x - \frac{\pi}{4}) \right], \quad (37)$$

$$\xi(x, 0) = \xi_0 + \varepsilon a \left[\frac{e^{k_l \eta}}{\sqrt{k_l}} \sin(k_l x - \frac{\pi}{4}) + \frac{e^{k_r \eta}}{\sqrt{k_r}} \sin(k_r x - \frac{\pi}{4}) \right], \quad (38)$$

where (η_0, ξ_0) is a numerically exact Stokes wave and $\varepsilon = 0.1$.

Results: three full recurrence cycles over 716 carrier periods. Energy conserved to $O(10^{-3}\%)$ for $t/T \leq 600$. $M = 2$ suffices (Benjamin–Feir is a four-wave process).

5.4 Post-NLS Effects

The full Euler equations exhibit phenomena absent from NLS:

- Recurrence breakdown after 2–3 cycles (higher-order resonances).
- Permanent frequency downshift ($O(\varepsilon^4)$ effect).
- Wave breaking for $k_0 a \gtrsim 0.3$ (the DNO ceases to be defined when η becomes multi-valued).

These are precisely the regimes where a learned approximation of the full DNO would be valuable: it captures the complete nonlinear map without truncation at finite M .

6 Generating Deep-Water Training Data

The preceding analysis provides five families of physically consistent (η, ξ) pairs in deep water. For each, one evaluates $G(\eta)\xi$ via the DNO series at $h = 1000$ (deep-water proxy).

Source	Parameters	Nonlinearity
Stokes waves	k_0, a_0	mild (single frequency)
Linear superposition	random spectrum	nearly linear, broad bandwidth
Envelope solitons	A_0, k_0	moderate (localized envelope)
BF trajectories	$k_0, a_0, \delta k, t$	strong (time-varying cascades)
Akhmediev/Peregrine	κ, A_0	extreme (up to $3\times$ background)

Linear superposition is cheapest: draw random k_j, a_j, ϕ_j and set

$$\eta(x) = \sum_j a_j \cos(k_j x + \phi_j), \quad \xi(x) = \sum_j \frac{a_j \omega_j}{k_j} \sin(k_j x + \phi_j), \quad \omega_j = \sqrt{g|k_j|}.$$

BF trajectories are most informative: time-integrate (1)–(2) from (37)–(38) and snapshot at intervals.

Depth conditioning. A model trained at $h = 1$ learns $G_0 = |k| \tanh(k)$ and cannot extrapolate to $G_0 = |k|$. For finite depths, the rescaling $x \rightarrow x/h$, $\eta \rightarrow \eta/h$ reduces to $h = 1$. The deep-water limit is genuinely different and requires either direct training data or a depth-conditioned architecture (e.g. encoding $\tanh(hk)$ as an input channel).

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